Tau Core: An Atemporal Morphological Readout Framework for Emergent Physical Descriptions

Foundation Paper I

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Abstract

Purpose. Tau Core is proposed here as a cautious foundation framework rather than a completed physical theory. Its starting point is the possibility that ordinary dynamical physics is not fundamentally time evolution on a primitive spacetime, but an effective readout of a deeper atemporal response structure.

Formal spine. The paper introduces the minimal formal spine

$$s \longmapsto \rho_\tau(s) \longmapsto U_i^{M_\tau}(\rho_\tau(s)),$$

where s is a parent-side seed, $\rho_\tau(s)$ is an endpoint-blind Tau response, M_τ is the Tau-side morphological body, and $U_i^{M_\tau}$ are morphology-conditioned sector readouts such as spacetime, observer-time, quantum, gravitational, or mass/source readouts.

Scope. The central claim is architectural. The formalism gives a language for separating definitions, postulates, conditional implications, toy mechanisms, and future empirical tests. The manuscript distinguishes the framework from block-universe ontology and from ordinary spacetime field theory, defines the Tau-side morphological body, formulates seven foundation postulates, introduces readout-relative hidden defects via relift failure, and gives finite toy models for hiddenness and effective readout-time. This manuscript should be evaluated as a foundation-language and claim-boundary paper, not as a phenomenological or empirical result paper.

Boundary. It also records the main blockers. The paper does not derive general relativity, quantum mechanics, the Born rule, quantum field theory, the Standard Model, dark matter, dark energy, or a physical parent action. It is a position paper and formal conjecture scaffold, not empirical evidence for a new physical law.

Manuscript Roadmap

1 Purpose and claim boundary

This manuscript is the first Roman-numbered Tau Core foundation paper. It is separate from the Arabic-numbered Tau Core paper sequence, where the latter is used for bounded empirical or protocol packages. The purpose here is more basic: to state the current foundation language in a form that can be criticized, compared, and refined. Until the operational papers are public or archived, this manuscript treats them as internal provenance only and does not rely on them as evidence.

The claim boundary is deliberately narrow. The paper proposes a formal architecture and audit discipline, not a validated physics model. It gives a vocabulary in which later questions about spacetime, quantum theory, gravity, and dark-sector language can be stated without assuming that spacetime time evolution is the primitive level. The main exclusions are summarized below and itemized again near the end of the manuscript.

> **Reviewer-facing claim box.** This paper claims a formal readout architecture and a failure discipline. It does not claim empirical validation, a new force, dark-matter replacement, or a derivation of GR/QM/Standard-Model physics. Promotion of any later branch requires source freeze, quotient safety, known-limit recovery, endpoint-blind prediction or exclusion, and survival under the relevant holdout/control test.

The central idea can be written in one line:

$$s \longmapsto \rho_\tau(s) \longmapsto U_i^{M_\tau}(\rho_\tau(s)). \tag{1}$$

The first arrow is not time evolution. It is a parent-side response relation. The second arrow is a readout. A physical theory appears only after a readout has supplied a codomain, a null policy, a closure test, and a validation boundary. When M_τ is fixed by the declared source class, the superscript is often suppressed and U_i is written for $U_i^{M_\tau}$.

In a deterministic branch one may write

$$\rho_\tau : \mathcal{S} \to \mathfrak{R}_\tau.$$

In a relational branch the same symbol denotes either a relation

$$\rho_\tau \subseteq \mathcal{S} \times \mathfrak{R}_\tau$$

or, equivalently, a set-valued response rule

$$\rho_\tau : \mathcal{S} \to \mathcal{P}(\mathfrak{R}_\tau).$$

Such a branch must declare whether it uses the whole admissible response class or an endpoint-blind source-freeze rule selecting a representative.

Equivalently, the paper works with the following foundation package:

$$\mathfrak{T} = (\mathcal{S}, \mathfrak{R}_\tau, \rho_\tau, M_\tau, \{U_i\}, \{N_i\}, \{J_i\}).$$

Here $\mathcal{S}$ is a seed/source domain, $\mathfrak{R}_\tau$ is a formal response domain, ρ_τ is a map, relation, or set-valued response rule, M_τ is the morphological body, U_i are sector readouts, N_i are null or quotient policies, and J_i are optional relifts. This package is an organizing object for later branches, not a proof that Nature uses it.

In informal Tau Core discussions, the phrase *Tau Core base field* refers to this parent-side response substrate, represented here by $(\mathcal{S}, \mathfrak{R}_\tau, \rho_\tau)$, together with the morphological body M_τ. The phrase is heuristic. In this paper it does not denote a physical field with an action, equations of motion, units, stress-energy, or empirical calibration.

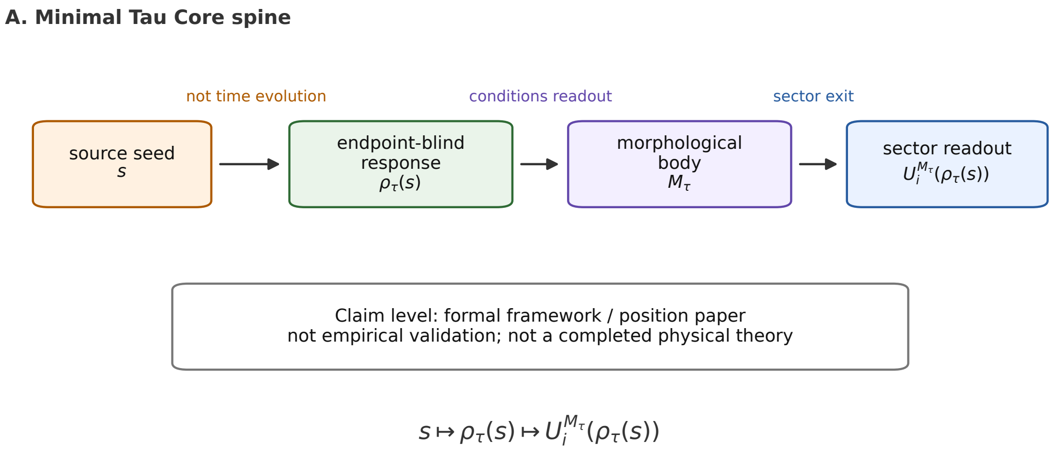

Figure 1: The minimal Tau Core spine used in Foundation Paper I. A parent-side seed s induces an endpoint-blind Tau response $\rho_\tau(s)$. A Tau-side morphological body M_τ conditions sector readouts U_i. This is a formal architecture, not empirical validation.

Method note. The manuscript was developed in an AI-assisted theoretical-research workflow. The AI workflow is not used as evidence for any physics claim; it is only part of the drafting and audit process.

This paper is therefore closer to a formal conjecture paper or a position paper than to a phenomenology paper. It is useful only if it makes the research program more falsifiable later. A broad ontology that cannot be turned into frozen source rules, null tests, or endpoint-blind predictions is not useful as physics. The framework below is written with that constraint in mind.

2 Reader roadmap

The paper has two jobs. First, it fixes a vocabulary for later Tau Core work. Second, it states what that vocabulary does not yet prove. The following roadmap is included to prevent the foundation language from being read as a completed physical model.

Part of the paper	Role	Claim level
Opening sections	Motivation, name, and relation to block-universe and spacetime-field views	Positioning
Core vocabulary	Parent response, morphological body, readout, relift, defect	Definitions
Formal spine	Minimal calculus, source discipline, effective readout-time, and postulates	Framework postulates
Toy layer	Finite hidden-defect model and morphology/observer-time clarifications	Toy mechanisms
Boundary layer	Quantum, gravity, cosmology, relation to earlier papers, and objections	Boundary setting
Audit layer	Maturity checklist, foundation sequence, compact summary, and non-results	Audit discipline

Thus the paper should be read from weak to strong claim levels. Definitions do not become assumptions; assumptions do not become derived laws; toy models do not become empirical validation. Later papers may try to promote specific branches, but Foundation Paper I keeps those promotions outside its claim boundary.

3 Why a foundation paper is needed

Several themes in modern foundations already question whether time, spacetime, or local dynamics should be primitive. Canonical quantum gravity contains the well-known problem of time, where a timeless constraint equation can coexist with effective dynamics [9, 25]. Relational time constructions show how dynamics can be described through correlations rather than external evolution [21]. Timeless or configuration-space views have also been explored philosophically and mathematically [2]. The thermal-time hypothesis is especially close in spirit because it treats time as state-dependent rather than as an external primitive [6]. Causal-set and generalized quantum-mechanics programs provide other examples of frameworks that change the primitive descriptive layer rather than merely adding a term to an equation [4, 14]. Quantum reconstruction programs show that large parts of quantum formalism can be organized by operational principles rather than simply postulated [13, 5, 18]. Constructor theory is a useful methodological comparison because it also tries to state physical content through possible/impossible transformations rather than through one specific dynamical equation [7, 8].

Tau Core does not claim to solve those programs. It uses a different organizing question:

> Can observed sector descriptions be treated as readouts of a deeper parent response, so that time evolution, quantum state, gravity, mass, and observer-time are effective sector descriptions rather than primitive categories?

This question is intentionally broader than any one branch. If it is too broad, it becomes metaphysics. The purpose of the present paper is to prevent that drift by specifying the minimal objects that must exist before a Tau Core statement becomes mathematically meaningful.

The immediate motivation is the following tension. Many physical descriptions use variables that are defined only after an observer, a section, a clock, a measurement basis, or a spacetime background is in place. But the same description often treats those variables as if they were primitive. Tau Core instead separates:

$$\text{parent response} \quad \text{from} \quad \text{sector readout.}$$

That separation is the entire point of the core-spine relation (1).

4 The name Tau Core

The name "Tau Core" is historical. The program began from a time-based intuition: perhaps the time seen by a four-dimensional observer is not a primitive background, but a projected response. This explains the use of τ, a symbol often associated with time or proper time in physics.

In the present architecture, however, τ does not mean ordinary time. It names the parent-state side of the theory. Time is one possible readout from that side. The distinction is important:

Symbol or phrase	Meaning here	Not claimed here
τ	parent-state side	ordinary time itself
X_τ	parent Tau state	detected physical field
$\rho_\tau(s)$	endpoint-blind response	time evolution
U_{time}	time readout	derived clock theory
M_τ	morphological body	observed 4D shape

Thus the theory is time-motivated but not time-reductionist. It does not say that everything is made of time. It says that time belongs to the readout side, while τ names the deeper side from which time and other physical sectors may be read.

5 Difference from a block universe

A block-universe reading treats a four-dimensional spacetime history as the final object. The comparison target is not a straw man. The relativistic background begins with Minkowski's four-dimensional spacetime formulation [20]. The Rietdijk–Putnam–Maxwell line of argument then uses relativity of simultaneity to motivate four-dimensionalism or eternalism [24, 23, 19, 28]. Standard surveys of the philosophy of spacetime and time describe the resulting debates around being, becoming, eternalism, and the block universe [26, 17, 22, 27]. In that setting one may summarize the block-universe reading as

$$\text{Reality} = M_{4D}. \tag{2}$$

Tau Core instead treats a four-dimensional block, when such a block is a good description, as a readout:

$$M_{4D} = U_{4D}^{M_\tau}(\rho_\tau(s)). \tag{3}$$

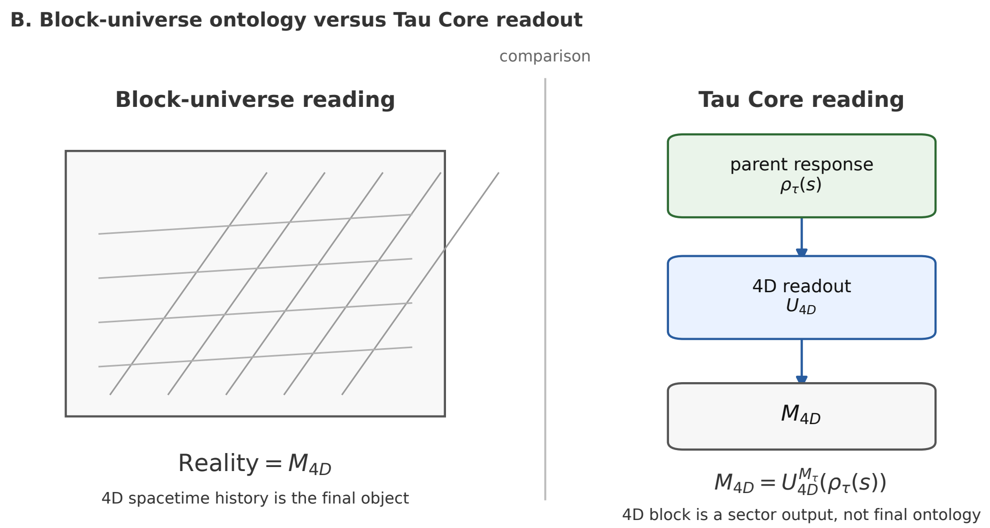

Figure 2: The block-universe comparison used in this paper. The left side represents the standard four-dimensional block reading, in which the spacetime history is the final object. The right side represents the Tau Core position: the same effective 4D block may be a stable readout $M_{4D} = U_{4D}^{M_\tau}(\rho_\tau(s))$ of a deeper parent response. The figure is a conceptual distinction, not a proof against eternalism.

This difference has physical content only if additional non-spacetime readout constraints can be source-frozen beyond the spacetime block itself. If no non-spacetime readout imposes such restrictions, the distinction remains a metaphysical rephrasing rather than a physical branch. The difference is not cosmetic. In a block-universe view the four-dimensional object is the final arena. In Tau Core it is an arena produced by a readout. The parent response can in principle have distinctions that one readout identifies, another ignores, and a third translates into effective readout-variation.

This distinction also clarifies why Tau Core is not simply another form of eternalism. It can be compatible with an effective 4D block where that description works, but it does not take the block as ontologically final. Nor does the paper refute the block-universe literature. It relocates the four-dimensional block from final ontology to sector output:

$$\text{block as final object} \quad \longrightarrow \quad \text{block as } U_{4D}\text{-readout.}$$

6 Difference from ordinary spacetime field theory

An ordinary field theory begins with a field on spacetime:

$$\phi : M_{4D} \to V, \tag{4}$$

and then studies the dynamics of $\phi(x,t)$. Tau Core rewrites the order:

$$\phi_{\text{obs}}(x,t) = U_\phi(\rho_\tau(s)). \tag{5}$$

The observed field is a readout. The spacetime coordinates are also part of the readout structure. This does not make ordinary field theory wrong. It makes it effective: a very successful description after a particular sector readout has become stable.

This is why Tau Core cannot claim physical success merely by writing a new readout map. To become physics, a readout must recover known limits and must survive empirical controls. If U_ϕ is arbitrary, the framework is empty. The research program therefore needs source-frozen readout rules, quotient-safe null policies, and explicit failure conditions.

7 Core vocabulary

Definition 1 (Parent seed). *A parent seed $s \in \mathcal{S}$ is a Tau-side source datum. It is not an observed endpoint and must not be chosen by looking at the endpoint residual that a later readout is supposed to explain.*

Definition 2 (Endpoint-blind Tau response). *An endpoint-blind Tau response is a map or relation*

$$\rho_\tau : \mathcal{S} \to \mathfrak{R}_\tau$$

whose input is the parent seed and admissible parent-side constraints, not the observational endpoint. Endpoint blindness is a non-leakage rule, not a proof that the response is physically real. When ρ_τ is relational rather than single-valued, $\rho_\tau(s)$ denotes the admissible response class, or a representative chosen by an explicitly declared source-freeze rule. No endpoint residual may be used to select the representative.

Definition 3 (Morphological body). *The Tau-side morphological body M_τ is the intrinsic structured body that conditions readouts. It is not an observed 4D morphology label. It may carry sector, branch, wall, null, Hessian-support, admissibility, history, and observer/path-relevant structure. A projected morphology is only a readout handle on this deeper body.*

For Foundation Paper I the minimal mathematical type is:

$$M_\tau \in \mathcal{M}_\tau^{\mathrm{adm}},$$

where $\mathcal{M}_\tau^{\mathrm{adm}}$ is a declared class of admissible morphological records. The required part of such a record is deliberately small: it must specify the source domain, the readout family, the null policy, and the admissibility/freeze rule used by the branch. Sector decompositions, walls, Hessian support, history, and observer/path structure are optional branch data unless the branch explicitly uses them in its readout rule. A branch may instantiate such a record schematically as

$$M_\tau = (S, B, W, N, H, A; \Theta),$$

where S records sector decomposition, B branch/support structure, W wall or transition adjacency, N null/protected structure, H Hessian-response support, A admissibility constraints, and Θ optional interface/history/path data. Thus $U_i^{M_\tau}$ is not merely a decorative superscript: it means that the readout map is selected from a typed family

$$U_i : \mathcal{M}_\tau^{\mathrm{adm}} \times \mathfrak{R}_\tau \to \mathcal{O}_i.$$

This is still a minimal type declaration, not a construction theorem for $\mathcal{M}_\tau^{\mathrm{adm}}$. In particular, an observed galaxy morphology, a quantum access structure, and an observer-time structure may be different projected handles on different subrecords of M_τ, not identical kinds of morphology.

Definition 4 (Readout)**.** *A readout is a morphology-conditioned sector map. Equivalently, one may write*

$$U_i^{M_\tau} : \mathfrak{R}_\tau \to \mathcal{O}_i, \qquad \text{or} \qquad U_i : \mathcal{M}_\tau \times \mathfrak{R}_\tau \to \mathcal{O}_i.$$

The codomain $\mathcal{O}_i$ may be a spacetime description, a time ordering, a quantum access state, a gravitational response, a mass/source record, or a more specialized observable structure. Each readout has its own null directions and validation boundary. When M_τ is fixed by the declared source class, we suppress it and write $U_i(r)$.

Readout family	Parent/source class	Readout form	Minimal freeze target
Spacetime	local chart and metric handles	$U_{4D}^{M_\tau}(r)$	local interval or chart limit
Observer-time	clock, interface, and path records	$U_{\text{time}}^{M_\tau}(r, \mathcal{O}, \gamma)$	proper-time limit or frozen path correction
Quantum access	access, phase, context, or outcome records	$U_q^{M_\tau}(r)$	normalized access probabilities or a null/control rule
Gravity	weak-field, closure, lensing, or acceleration records	$U_g^{M_\tau}(r)$	Newtonian/GR limit or quotient-visible residual
Mass/source	density, source-measure, flavor, or particle-content records	$U_m^{M_\tau}(r)$	conserved source measure or declared mass/source map
Cosmology	web, void, path, clock, or expansion records	$U_{\text{cosmo}}^{M_\tau}(r)$	FLRW/ΛCDM limit or frozen residual channel

Table 1: Readout families and illustrative parent/source classes. The table does not derive the families physically; it records the minimal bookkeeping needed before a branch can become more than a vocabulary. The freeze targets are intentionally heterogeneous placeholders. Each later branch must replace them by operationally defined quantities with units, uncertainties, null models, and failure rules; they are not yet calibrated observables.

Definition 5 (Cross-scale readout composition)**.** *Readout families are branch-distinct but not assumed independent. A candidate kernel K_i for one family is admissible only after the source, carrier, quotient, and observable dependencies that compositionally link it to neighboring readout families have been checked, or explicitly frozen as negligible under the declared assumptions. Equivalently, Tau Core treats physical readouts as a compatibility-coupled composition hypergraph, not as isolated fit modules.*

This is an admissibility principle, not a proof of unification. It does not derive quantum gravity, particle physics, general relativity, or cosmology. Rather, it forbids treating, for example, a galaxy-scale gravity kernel as independent of the coarse-grained particle, mass, material, radiation, and path readouts that compose the observed carrier. It also prevents double counting: two source-overlapping readout terms cannot be added as independent corrections unless their non-overlap and quotient visibility have been declared.

Definition 6 (Relift)**.** *A relift for a readout U_i is a map or partial map*

$$J_i : \mathcal{O}_i \to \mathfrak{R}_\tau$$

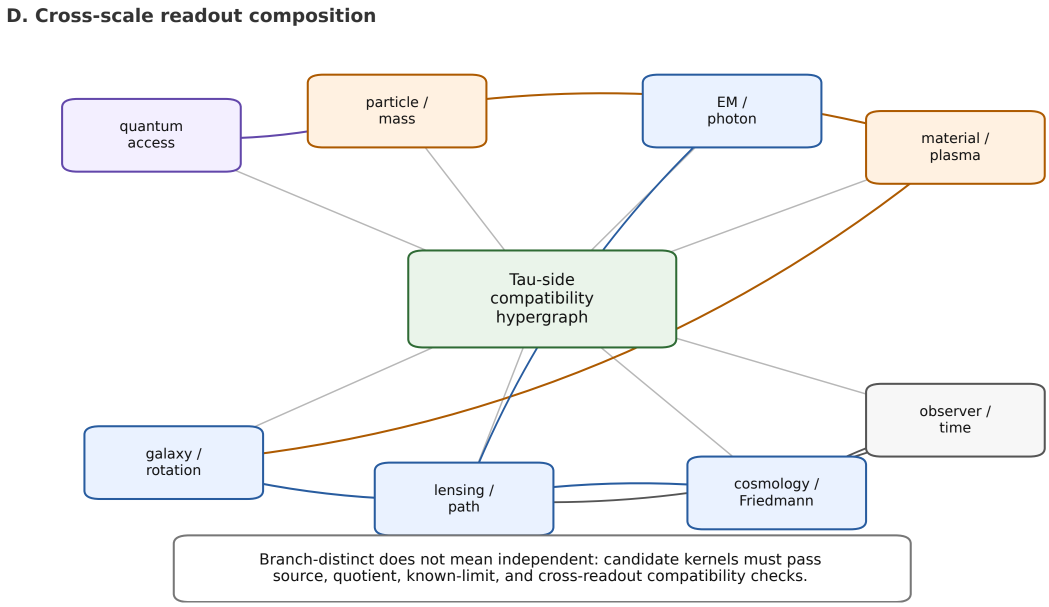

Figure 3: Cross-scale readout composition. Tau Core readout families are branch-distinct, but branch distinction does not imply operational independence. Candidate kernels must respect source, carrier, quotient, and cross-readout compatibility constraints before they can be promoted beyond scaffold status. The figure is architectural: it is not a proof of physical unification.

that attempts to reconstruct a parent-response representative from a readout record. It need not exist globally and need not be unique.

Definition 7 (Readout obstruction)**.** *Given $r = \rho_\tau(s)$, the primary defect object is not a subtraction but an obstruction record*

$$\omega_i(r) = 0 \quad \Longleftrightarrow \quad J_i U_i(r)\, N_i\, r,$$

whenever a relift and null relation have been declared. Thus $\omega_i(r) \neq 0$ records that the readout and its allowed relift fail to recover the parent-response representative modulo the declared null relation. Only in a chosen affine or linearized chart may this be represented by the coordinate expression

$$h_i(r) = J_i(U_i(r)) - r.$$

The obstruction condition and its chart-level representative are not dark-matter formulas or quantum-collapse formulas. They are abstract incompleteness records. A physical branch must later specify the codomain, quotient, units, known limits, and empirical test.

8 Status of the primitive objects

A common failure mode for foundation papers is to let definitions, assumptions, and derived claims blur into each other. The following ledger records the status of the new objects used above. It is not additional evidence; it is a claim-boundary device.

Primitive-object status ledger

Object	Status in this paper	Not supplied here
$\mathcal{S}$	declared source/seed domain	a physical ontology of all sources
$\mathfrak{R}_\tau$	parent-response space, assumed as formal domain	uniqueness or physical existence proof
ρ_τ	endpoint-blind response map or relation	dynamical law or parent action
M_τ	morphology-conditioning body	derivation from first principles
$U_i^{M_\tau}$	sector readout definition	a calibrated physical observable model
N_i	declared null/quotient policy	automatic gauge principle
J_i	optional relift device	globally defined inverse readout
ω_i	obstruction record modulo N_i	dark-sector or collapse mechanism

The paper becomes nontrivial only if later branches replace entries in the third column by source-side derivations, known-limit recoveries, or failed controls. Until then these objects define a disciplined language, not a completed physical theory.

9 Minimal readout calculus

The formalism above is intentionally weak. A readout is not yet a functor, a field equation, a probability rule, or a variational principle. Nevertheless, some minimal calculus is needed if the framework is to avoid being only language.

The mathematics in this section is standard. A readout induces a partition of the parent-response space; quotient safety is well-definedness on the corresponding quotient; refinement is the ordinary refinement order on partitions. Tau Core does not claim novelty for these facts. It uses them as bookkeeping discipline so that later physical branches cannot hide endpoint leakage or double counting behind new terminology. The same conceptual territory overlaps with coarse-graining, marginalization, consistent-history and decoherence-inspired languages, and Jaynes-style information-theoretic statistical mechanics [12, 10, 15, 16].

Let $\mathfrak{R}_\tau$ be a parent-response space and let U_i be a readout. The first mathematical object induced by a readout is its indistinguishability relation:

$$r \sim_i r' \iff U_i(r) = U_i(r'). \tag{6}$$

This relation is readout-relative. A distinction can be invisible to one sector and visible to another. In this sense, a readout is not just a map to observables; it is also a rule for what counts as the same state for that sector.

Definition 8 (Readout kernel)**.** *The readout kernel of U_i is the equivalence relation $\sim_i$ defined by the readout-equivalence relation* (6)*. If $\mathfrak{R}_\tau$ has a linear structure and U_i is linear, this reduces to the usual null space* $\operatorname{Ker} U_i$*. In general it is an equivalence relation, not necessarily a linear subspace.*

Definition 9 (Readout refinement)**.** *A readout U_j refines U_i if*

$$r \sim_j r' \Longrightarrow r \sim_i r'.$$

Equivalently, U_j distinguishes at least as much parent-response information as U_i.

This gives a minimal partial order on readouts. A more informative readout is not automatically more physical. It may be harder to stabilize, harder to observe, or more vulnerable to noise. But the refinement relation prevents one from treating all readouts as interchangeable.

Definition 10 (Joint readout)**.** *Given readouts U_i and U_j, the joint readout is*

$$U_{ij}(r) = (U_i(r), U_j(r)).$$

Its equivalence relation is the intersection:

$$\sim_{ij} = \sim_i \cap \sim_j .$$

Proposition 1 (Joint readouts refine their components)**.** *The joint readout U_{ij} refines both U_i and U_j.*

Proof. If $r \sim_{ij} r'$, then $U_i(r) = U_i(r')$ and $U_j(r) = U_j(r')$. Hence $r \sim_i r'$ and $r \sim_j r'$. Therefore the joint equivalence relation is contained in each component equivalence relation. $\square$

This trivial observation is one reason multi-readout tests are valuable, but it should not be confused with statistical independence. Two readouts may be correlated sibling readouts of the same parent. A minimal observation model is

$$R_i = O_i[P] + N_i,$$

where P is a shared latent parent, O_i is the sector readout operator, and N_i collects noise, selection, and pipeline effects. In cosmology, for example, galaxy clustering, weak lensing, gas tracers, and CMB lensing can all respond to the same matter/potential parent while using different observational operators.

Therefore raw agreement between readouts is not, by itself, a Tau Core signal. The stronger question is whether a common residual remains after conditioning on the appropriate null parent P_0:

$$\Delta R_i = R_i - O_i[P_0].$$

For a cosmological branch P_0 might be a matched ΛCDM matter/potential parent; for another branch it would be the relevant standard baseline. A Tau Core-compatible excess would be residual cross-readout structure that survives the declared null model, covariance, and source-overlap ledger. Here this is simply called source-overlap discipline. It is not a validation statistic and not an independence claim. Without this discipline, a joint readout may simply double count the same source coordinate.

Unpublished Tau Core protocol notes use this same discipline in more specific empirical settings, but those notes are not evidence for this foundation paper. They are therefore not used here as supporting results.

10 Nontrivial representation condition

The parent/readout notation is not allowed to count as a Tau Core result by itself. Without an extra restriction, any effective theory could be written in the trivial form

$$\mathfrak{R}_\tau = \mathcal{O}, \qquad \rho_\tau(s) = O, \qquad U = \mathrm{id}_{\mathcal{O}}.$$

Such a representation is formally possible but scientifically empty.

Foundation Paper I therefore treats the following as a nontriviality requirement rather than as a theorem:

> A promoted Tau Core branch must use a source-declared parent response and a readout family that cannot be reduced, without loss of the declared null, source, or known-limit structure, to the identity readout on the endpoint observable.

Equivalently, the parent structure must do at least one of the following before the endpoint is inspected: impose a quotient-safe null policy, relate two or more readouts through a shared source coordinate, recover a known physical limit, or exclude a wrong-family/control construction. Otherwise the branch is only a relabeling of the endpoint theory.

Known-limit recovery is not sufficient by itself. Many harmless rewritings of an existing theory can reproduce an already-known limit. Promotion therefore also requires at least one endpoint-blind consequence that could have failed: a frozen numerical value, a forbidden family, a null prediction, a source-side relation, or a control exclusion declared before endpoint inspection.

11 Nulls, quotients, and protected identity

Every physical readout has null directions: changes in the parent response that the readout treats as physically irrelevant. Some nulls are ordinary gauge redundancies. Some are observational limitations. Some are protected identity relations: distinctions that should not be counted as physical differences in a declared sector.

Let N_i denote the null relation for readout U_i. The quotient object is written

$$[r]_i \in \mathfrak{R}_\tau / N_i.$$

The readout is quotient-safe only if

$$r N_i r' \Longrightarrow U_i(r) = U_i(r'). \tag{QS}$$

Equivalently,

$$N_i \subseteq \sim_i, \qquad U_i = \bar{U}_i \circ q_i, \qquad q_i : \mathfrak{R}_\tau \to \mathfrak{R}_\tau / N_i,$$

for an induced quotient readout $\bar{U}_i : \mathfrak{R}_\tau / N_i \to \mathcal{O}_i$. If the quotient-safety condition (QS) fails, the readout assigns different observables to parent states that it also declares physically equivalent. Such a readout is internally inconsistent.

The converse is not required. A readout can identify more states than the declared null relation identifies. That is precisely readout loss. The important issue is whether the loss is allowed. In empirical branches this is handled by a source-freeze discipline:

the readout-relevant source coordinates must be declared before endpoint scoring. In the foundation branch this becomes a source-priority problem: the selected readout should not erase protected source identity unless the erased distinction is explicitly routed to the null quotient.

A later formulation of the same idea is the readout-basin stability principle. For a declared 4D readout R_{4D}, a stable object-class O may be represented by the basin

$$\mathcal{B}_O = \{r \in \mathfrak{R}_\tau : R_{4D}(r) \simeq O\}.$$

Parent-response variation inside $\mathcal{B}_O$ preserves the observed object-class, while a visible transition or decay corresponds to crossing into another basin. In Foundation Paper I this is only a definition scaffold for object persistence under non-injective readout; it is not a proton-stability proof, a Standard Model derivation, or a decay-rate formula.

Definition 11 (Protected source identity)**.** *A protected source identity is an equivalence relation Q_{src} on parent-side seeds or source representatives that the theory treats as source-level identity. A readout rule is source-faithful if it does not collapse distinct Q_{src}-classes into one selected source class without declaring the collapse as protected null structure.*

This definition is not an extra empirical claim. It is a consistency demand. Without it, a model can hide arbitrary distinctions in the parent space and recover them only after seeing an endpoint. That would make the theory unfalsifiable.

12 Endpoint blindness as a scientific rule

Endpoint blindness is the operational face of the foundation idea. A readout model is endpoint-blind when the source, kernel family, amplitude rule, and null policy are fixed before the target endpoint residual is inspected.

This is not a philosophical preference. It is the only way a broad readout framework can avoid becoming curve fitting. If a galaxy rotation residual, a cosmological distance residual, or a quantum outcome is used to choose the source class that then explains it, the model has leaked the answer into its input.

The minimal endpoint-blind record contains:

(E1) a source object or source proxy;

(E2) a declared readout family U_i;

(E3) a null and quotient policy;

(E4) a source-overlap decision for all channels used together;

(E5) a freeze artifact or equivalent timestamped declaration;

(E6) forbidden endpoint fields;

(E7) controls and failure conditions.

Foundation Paper I does not run such an endpoint test. It states why later branches need it. The philosophical content of Tau Core becomes scientific only when it produces source-frozen rules that can fail.

13 How effective readout-variation appears

Tau Core replaces primitive time evolution by readout-parametrized variation. Let $r = \rho_\tau(s)$ be atemporal at the parent-response level. Suppose a readout U_i produces a record that can be coordinatized by a parameter t_i:

$$U_i(r) = o_i(t_i).$$

Then the derivative

$$\frac{do_i}{dt_i}$$

is meaningful inside the readout. It need not correspond to a parent-time derivative. This is the minimal sense in which this paper uses the phrase "effective dynamics": a parameterized variation internal to a readout, not a derived physical equation of motion.

The distinction can be made sharper by considering two readouts of the same parent response. One readout may admit a temporal parameter t_A, while another admits a static relational record:

$$U_A(r) = o_A(t_A), \qquad U_B(r) = C_B.$$

There is no contradiction. The first readout organizes the parent response as a time-indexed sequence. The second organizes it as a correlation or constraint. A conventional physical theory may then choose the first description because it is useful, local, and predictive. Tau Core's claim is weaker: usefulness of a temporal parameter does not make time primitive.

This is also why observer time must be sector-typed. A clock used in a galaxy-rotation analysis, a lensing time-delay analysis, a cosmological distance analysis, and a quantum measurement protocol are not automatically the same parent object. They may share a local proper-time limit, but that limit must be derived or justified inside the declared branch.

14 Toy model with effective readout-time

The finite model above can be extended to show effective readout-time. Let

$$r = (a_0, a_1, \ldots, a_n, c) \in \mathbb{R}^{n+2}.$$

Define a readout

$$U_{\text{seq}}(r) = (a_0, a_1, \ldots, a_n),$$

and choose the index k as an effective readout-time coordinate. The readout sees a sequence a_k. A finite-difference law can be defined:

$$\Delta a_k = a_{k+1} - a_k.$$

Nothing in the parent object has evolved. The parent response already contains the indexed structure. The readout has supplied an ordering and therefore an effective readout-variation.

Now define a second readout

$$U_{\text{sum}}(r) = \sum_{k=0}^{n} a_k.$$

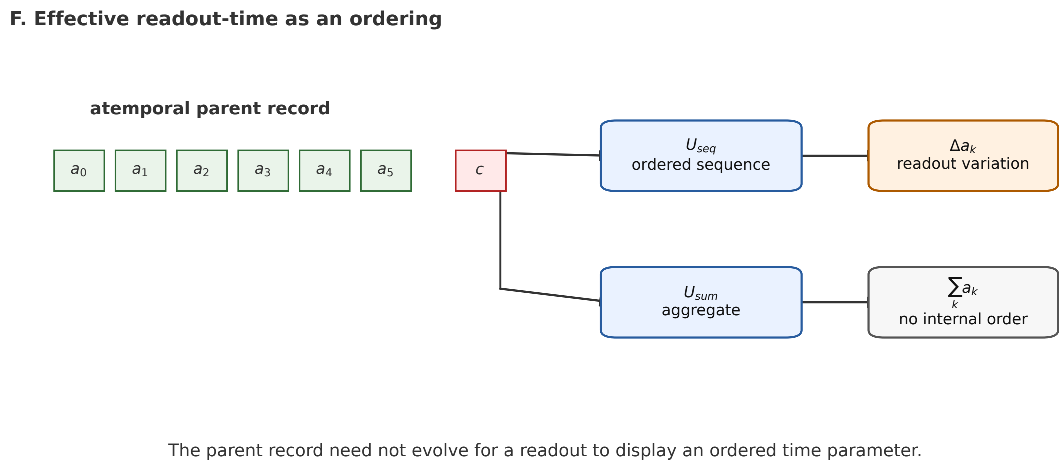

Figure 4: A finite toy model for effective readout-time. The parent record is not itself evolving. The sequence readout U_{seq} exposes an ordered list and therefore supports finite differences Δa_k. The aggregate readout U_{sum} exposes a different observable and loses the internal ordering. Effective readout-variation is therefore readout-relative in this model.

This readout sees no time sequence, only an aggregate. The hidden component relative to U_{sum} is the loss of ordered information. The hidden component relative to U_{seq} includes c. Again, hiddenness is readout-relative.

This toy model is deliberately too simple to be physics. It only shows that the sentence "time is a readout" can be made mathematically literal without introducing backward causation or a second physical clock.

A slightly richer finite time-readout toy, developed in the later Tau Core time-readout module, replaces the indexed vector by a small parent morphology graph. Candidate event orders can then be scored both by a local sequential path cost and by a global parent-consistency cost. In that toy the order that looks best as a local time story need not be the same as the order that best preserves parent affinities. This strengthens only the foundation-language point: order-sensitive narratives need not automatically be read as future-to-past influence. They can be treated, at toy level, as different time-order readouts of one globally constrained parent record. No physical clock theory, retrocausality, Born rule, or quantum dynamics is derived from this observation.

15 Seven foundation postulates

The postulates below are not proved in this paper. They define the candidate framework. Later papers or theory notes may try to derive or weaken them.

Postulate 1 (Tau substrate)**.** *There exists a parent-level structure $\mathcal{T}$ that is not, at the foundation level, ordinary spacetime evolution.*

Postulate 2 (Seed-response)**.** *Physical configurations are represented at the parent level by endpoint-blind responses to seeds:*

$$s \in \mathcal{S}, \qquad r = \rho_{\mathcal{T}}(s) \in \mathfrak{R}_{\mathcal{T}}.$$

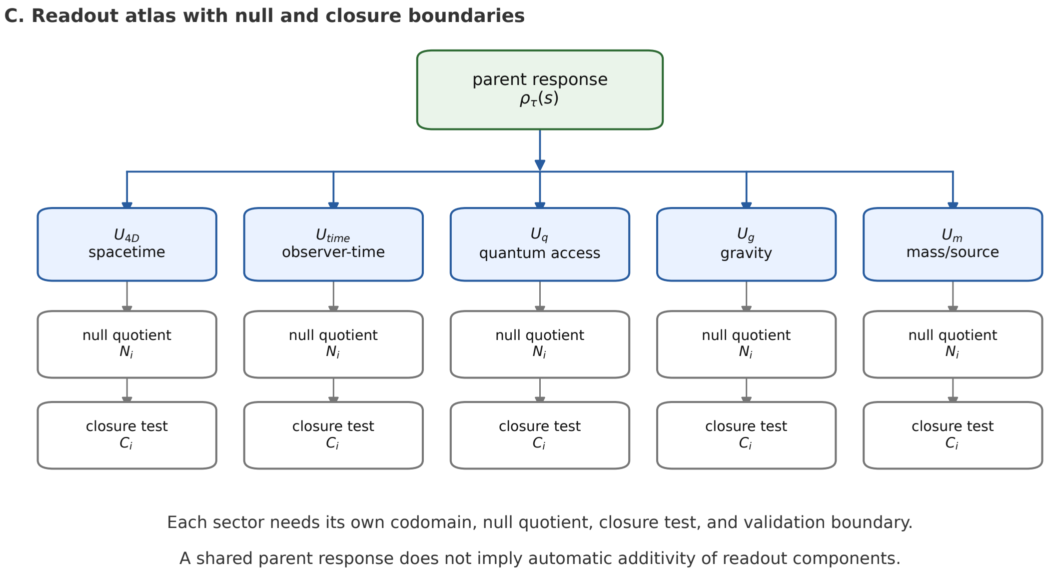

Figure 5: A parent response can have several sector readouts. The same parent object may be read as spacetime, observer-time, quantum access, gravitational response, or mass/source structure. Each sector requires its own codomain, null policy, closure test, and validation boundary. The paper does not assume that these readouts are automatically additive or mutually complete.

Postulate 3 (Endpoint blindness). *The response $\rho_\tau(s)$, and any source object used to select a readout kernel, must be specified without using the endpoint residual that the readout is later asked to explain.*

Postulate 4 (Readout emergence). *Observed physical descriptions are sector readouts:*

$$O_i = U_i(\rho_\tau(s)).$$

Time, spacetime, quantum access, gravity, and mass/source descriptions are readout candidates rather than primitive categories.

Postulate 5 (Effective readout variation). *Time evolution appears only when a readout admits a stable ordering or clock coordinate. A derivative such as*

$$\frac{d}{dt}U_i(\rho_\tau(s))$$

is an effective readout derivative, not a primitive parent-time derivative and not, by itself, a physical law of motion.

Postulate 6 (Hidden defect). *If a readout and its allowed relift fail to recover the parent-response representative modulo the declared null relation, the obstruction $\omega_i(r) \neq 0$ is recorded as a hidden defect of that readout. A coordinate difference $J_iU_i(r) - r$ is only a chart-level representative of that obstruction.*

Postulate 7 (Tension as readout failure). *Some observational or theoretical tensions may be modeled as readout, closure, or relift defects. This is a research hypothesis. It becomes physical only after source-frozen rules, controls, known limits, and held-out tests are supplied.*

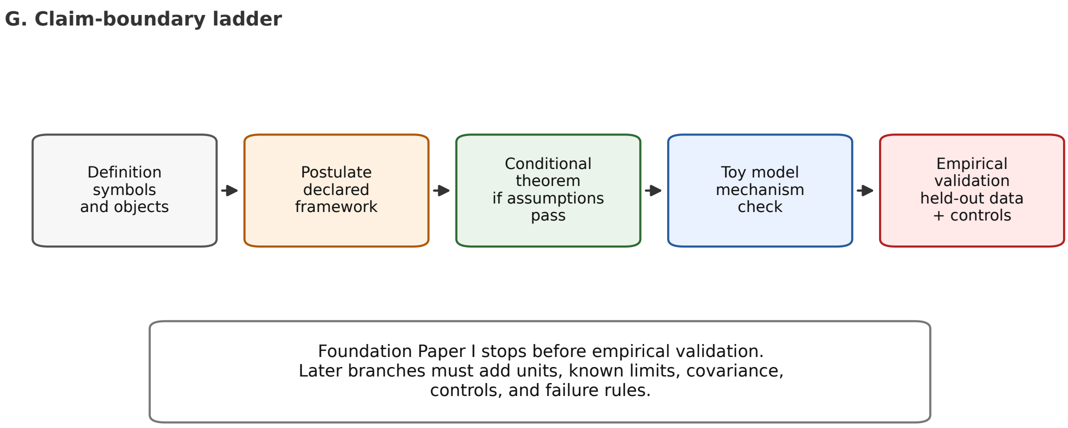

Figure 6: The claim ladder for Foundation Paper I. The present manuscript stops at definitions, postulates, conditional implications, and toy mechanisms. It does not claim empirical validation.

16 The claim ladder

A central discipline of Tau Core is to separate claim levels. The framework contains definitions, postulates, conjectures, conditional theorems, toy models, numerical evidence, and empirical validation. These are not interchangeable.

The most common failure mode for broad foundation programs is promotion by language: a metaphor becomes a mechanism, a mechanism becomes evidence, and evidence becomes validation. This paper avoids that promotion. For example, saying that observer-time is a readout does not derive a physical clock. Saying that dark-sector language may be a hidden-defect language does not remove dark matter. Saying that quantum state may be a readout does not derive Hilbert space.

A later Tau Core methodology note, the Tau Core Readout Eligibility Protocol v0.1 (TC-REP-0.1), packages this discipline as a pre-formal readout-eligibility audit. It asks whether a candidate branch excludes the target ontology from its parent assumptions, records primitive debt, passes through readout mediation, extracts a constraint, lands mathematically and operationally, separates from rivals, and preserves explicit failure modes. It is not a new mathematics or a proof by notation; it is a claim-boundary checklist for deciding whether a branch is more than renamed known physics. The protocol also admits a compact symbolic chain for audit cards, but this paper deliberately uses the prose checklist rather than treating that chain as derivation notation.

Figure 7 gives the same discipline in a more compact form. Green entries indicate what this paper supplies as framework content. Orange and red entries indicate explicit non-claims and blockers. The point is not to make the paper look complete; it is to make incompleteness visible.

H. Foundation Paper I claim-status matrix

	Defined	Postulated	Toy mechanism	Validated	Open blocker
Tau substrate	yes	yes		not claimed	open
Seed-response map	yes	yes		not claimed	open
Morphological body	yes	yes		not claimed	open
Sector readouts	yes	yes	yes	not claimed	open
Effective readout-time	yes	yes	yes	not claimed	open
Hidden defect	yes	yes	yes	not claimed	open
GR / quantum / QFT				not claimed	open
Empirical validation				not claimed	open

Green cells mark framework content supplied here; orange/red cells mark explicit non-claims or blockers.

Figure 7: Claim-status matrix for Foundation Paper I. The paper defines and postulates a parent-response/readout architecture and supplies toy mechanisms for effective readout-time and hidden defects. Orange and red entries mark the non-claims: no empirical validation and no GR/QM/QFT or parent-action derivation.

17 A finite toy model

The following model is intentionally small. Its role is not to reproduce physics. Its role is to show the readout logic without infinite formal machinery.

Let the parent response space be

$$\mathfrak{R}_\tau = \mathbb{R}^3, \qquad r = (a, b, c).$$

Let two readouts be

$$U_A(a, b, c) = (a, b), \qquad U_B(a, b, c) = (a, c).$$

These readouts forget different coordinates. Choose canonical relifts

$$J_A(x, y) = (x, y, 0), \qquad J_B(x, z) = (x, 0, z).$$

Since this toy response space is linear, $\mathfrak{R}_\tau = \mathbb{R}^3$, the obstruction can be represented by coordinate differences. This is a special toy-chart convenience, not the general definition of ω_i. Then the readout defects are

$$h_A(r) = J_A U_A(r) - r = (0, 0, -c),$$

and

$$h_B(r) = J_B U_B(r) - r = (0, -b, 0).$$

The same parent response therefore has different hidden components under different readouts. There is no contradiction: each readout has a different null policy.

Proposition 2 (Readout-relative hidden component)**.** *In the toy model above, a component hidden under readout U_A can be visible under readout U_B, and conversely.*

Proof. The coordinate c is absent from $U_A(a, b, c)$ and appears as the defect $(0, 0, -c)$ under the canonical relift J_A. The same coordinate is present in $U_B(a, b, c)$. Similarly, b is visible under U_A and hidden under U_B. Thus hiddenness is readout-relative. □

The lesson is small but important. A hidden component is not necessarily a new substance. It may be a component that a chosen readout cannot relift. Whether this idea has anything to do with dark sectors, quantum measurement, or cosmology is a later physical question. The toy model only licenses the logical grammar. In particular, it does not force a dark-sector mechanism, quantum mechanism, or gravitational correction.

Quotient-safety in the toy model. For U_A, the natural null relation is

$$(a, b, c)\, N_A\, (a', b', c') \quad \Longleftrightarrow \quad a = a' \text{ and } b = b'.$$

Then U_A is quotient-safe, because rN_Ar' implies

$$U_A(r) = U_A(r').$$

The coordinate c is invisible to U_A, but it is not therefore a new force or a new substance. It is simply a null coordinate for that readout. If one later modifies the same readout by adding

$$U_A^{\text{bad}}(a, b, c) = (a, b + \alpha c),$$

while keeping N_A as the declared null policy, quotient-safety fails: two N_A-equivalent parent representatives can now produce different observables. A branch that wants to make c physically visible must change the readout and null policy before endpoint inspection; it cannot silently use a null coordinate as an endpoint-saving correction.

Source-freeze in the toy model. Suppose a branch proposes a source rule

$$s = (a, b, c), \qquad c = f_{\text{src}}(a, b; \theta),$$

where f_{src} and θ are fixed from source-side information before any endpoint residual is inspected. Then $U_A(s)$, $U_B(s)$, and the corresponding defects are legitimate toy outputs of the frozen branch. If instead f_{src}, θ, or the decision to use c is chosen because U_A missed a target value, the branch has leaked endpoint information into the source definition. It may still be a phenomenological fit, but it is no longer a Tau Core source-frozen derivation.

18 Morphological body versus projected morphology

The phrase "morphology" can be misleading if left unqualified. In ordinary astronomical use it often means visible shape or catalogue type. In Tau Core, the primary object is not the observed shape. It is the Tau-side morphological body M_τ.

The morphological body is the structured source object that conditions readouts. A galaxy image, a rotation curve, a cosmic-web tensor, a lensing map, or a quantum measurement context can be a projected handle on the body, but not the whole body. This distinction became necessary because a single 4D appearance can underdetermine the readout-relevant source. For example, a present projected shape may not contain enough information about history, observer/path geometry, closure class, or null structure.

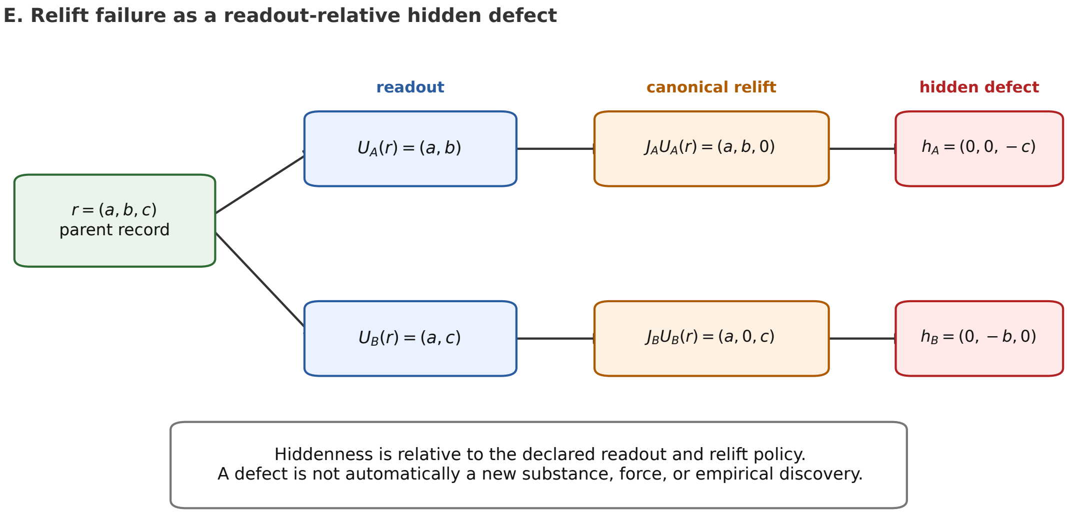

Figure 8: Readout-relative hidden defects. Two readouts of the same parent response can have different relift failures. A hidden component is not automatically a new force or substance; it is first an incompleteness record relative to a declared readout.

For Foundation Paper I, the minimal role of M_τ is:

$$O_i = U_i^{M_\tau}(\rho_\tau(s)). \tag{7}$$

The superscript indicates that the readout map is selected or conditioned by the morphological body. Equivalently one may regard U_i as a map on $\mathcal{M}_\tau \times \mathfrak{R}_\tau$. This does not say that M_τ has been physically derived. It says that readout selection is not source-free.

For a concrete branch, M_τ must also be admissible and frozen. This means that the projected handles used to select M_τ, the allowed family of readout kernels, and any source-overlap decisions between channels must be declared before endpoint scoring. If M_τ is changed after seeing the endpoint residual, the branch is phenomenological-only unless an independent source review reopens the selection.

This point also prevents double counting. If two proposed correction terms use the same source coordinate, they should not be added independently unless a source-side split justifies the separation. Otherwise the readout model is only relabeling the same source information twice.

19 Observer time

Tau Core treats observer time as a readout, not as a primitive parameter on the parent side. The schematic form is

$$t_{\text{obs}}^S = U_{\text{time}}^S(\rho_\tau(s), M_\tau, \mathcal{O}_S, \gamma_S), \tag{8}$$

where S is a sector, $\mathcal{O}_S$ is observer-interface data, and γ_S denotes path or section data relevant to that sector.

Equation (8) is not a derived clock theory. It is a discipline rule. It says that an observer's time belongs to the observer's readout bundle. A later physical theory must explain when t^S_{obs} reduces to proper time, when it admits corrections, and how those corrections can be measured without circularity.

This also clarifies why future-directed or history-like terms in Tau Core should not be read as backward causation. They are components of a deeper parent response whose different 4D readout slices may appear as past, present, or future. The parent object is not evolving in an external time. The observed ordering is created inside the readout.

20 Quantum readout

The quantum branch is one of the natural targets for the readout language, but it is also one of the most dangerous places to overclaim. A minimal quantum readout might have the form

$$q = U_q(\rho_\tau(s)), \tag{9}$$

where q is an access-state or probability-assignment record. But this is far below deriving Hilbert space or the Born rule.

Existing reconstruction programs show that quantum theory can be approached from information-theoretic or operational axioms. Examples include Hardy-style, purification-style, and local/reversible reconstruction routes [13, 5, 18]. Generalized probabilistic and Jordan-algebraic routes clarify how much structure is needed before Hilbert-space quantum theory is forced [3]. Tau Core can use these results as boundary discipline. It must not claim that a finite contextual readout grammar already implies complex Hilbert space.

The current safe statement is:

> If Tau Core supplies an ordered convex access-state space, separating effects, an appropriate self-dualizing pairing, continuous homogeneous reversible symmetry, composition rigidity, and trace uniqueness, then a Born/Hilbert representation becomes a conditional target.

The list is included to orient the reader, not to smuggle in quantum mechanics. Convex states and separating effects encode preparation and measurement separation; symmetry, composition, and trace conditions are the kinds of regularity assumptions that distinguish Hilbert-like quantum theory from a generic probability calculus. Tau Core would still have to derive or independently justify any such assumptions in a later quantum branch. The conditions listed in this sentence are not postulates of Foundation Paper I. They are examples of the additional reconstruction structure that a later quantum branch would have to supply or replace.

One possible future worked branch is a finite phase-gluing toy model. In such a model one would specify selected overlaps, transition phases Γ_{ij}, and a cocycle-like consistency condition

$$\Gamma_{ij} + \Gamma_{jk} + \Gamma_{ki} = 0 \mod 2\pi$$

on triples before claiming a global phase chart. This resembles the ordinary mathematics of gluing local phase data; it is not a Tau Core proof. Only after the finite domain, overlap cover, null policy, trace/effect pairing, and source-freeze rule are stated could such a toy branch be assessed. Foundation Paper I therefore does not claim a Hilbert/Born recovery. It only identifies what a later quantum branch would have to prove.

In this limited sense, the quantum readout is not a sealed sector. A parent-side distinction with access, phase, contextual, or noncommuting structure may have quantum-facing and non-quantum-facing projections. If such a distinction survives the null quotient of a gravity, mass/source, observer-time, or cosmological readout, a quantum-like signature could appear inside that sector's residual structure. This is not the claim that quantum mechanics acts directly on every other readout. It is the weaker claim that several readouts may expose different aspects of the same parent-side structure, and that any shared use must be controlled by source-overlap and double-counting rules. This is a conditional reconstruction route, not a derivation. The present paper includes quantum readout only to show where it sits in the common architecture.

21 Gravity and dark-sector readouts

The gravity branch motivates the hidden-defect vocabulary. A gravitational readout may fail to relift all source-relevant structure into a Newtonian, relativistic, or closure-compatible description. In abstract notation:

$$\omega_g(r) = 0 \quad \Longleftrightarrow \quad J_g U_g(r)\, N_g\, r, \qquad r = \rho_\tau(s). \tag{10}$$

If $\omega_g(r) \neq 0$ and the obstruction record survives the declared null quotient with a measurable 4D readout, it becomes a candidate residual channel. A coordinate difference

$$h_g = J_g U_g(r) - r$$

may be used only after choosing a linearized gravitational chart in which the subtraction is meaningful. This is still not a dark-matter replacement. It is a source of possible residual structure.

The same discipline applies to dark-sector language. Tau Core may eventually ask whether some dark-sector phenomena are better understood as readout-relative hidden defects. Foundation Paper I does not answer that question. It only defines the level at which the question becomes meaningful. To make a dark-sector claim, one would need source-frozen kernels, conventional baseline comparisons, covariance/systematics treatment, and failure criteria.

22 Cosmological readout

Cosmology is another natural readout target because the observed expansion history, distance ladder, growth data, lensing data, and CMB inference are all compressed through model-dependent readout assumptions. A Tau Core cosmological readout might be written

$$O_{\text{cosmo}} = U_{\text{cosmo}}(\rho_\tau(s), M_\tau^U, \gamma_{\text{path}}, \mathcal{O}_{\text{obs}}), \tag{11}$$

where M_τ^U is a universe-scale morphological body, γ_{path} is path/environment data, and $\mathcal{O}_{\text{obs}}$ is observer-interface data.

Again, the notation is not a result. It is a warning against compressing all cosmological tensions into a single function of the scale factor. A real cosmological Tau Core branch would need multi-probe covariance, source-native void/web/path/time channels, and a clear relation to ΛCDM limits.

23 Relationship to the existing paper sequence

The Arabic-numbered Tau Core papers are not required for the logic of this foundation paper. They are narrower, operational packages. Some examine galaxy residuals, some define source-frozen morphology or projection kernels, some formulate dark-sector decision protocols, and some define cosmological or web-witness audit ladders. They are evidence-seeking packages, not proofs of the parent ontology. Unless and until a given paper is public or otherwise archived, it should be read here only as internal provenance for terminology, not as a cited result.

Foundation Paper I has the opposite role. It defines the vocabulary that those operational papers can later use. It should not borrow their numerical outcomes as proof. A positive empirical audit in a later branch can motivate the framework, but it does not validate the parent theory unless the path from parent source to readout rule is itself frozen, non-circular, and tested.

This separation is important for scientific criticism. A critic should be able to reject a galaxy-kernel result without thereby rejecting the formal readout architecture. Conversely, a critic can reject the foundation architecture while still accepting that a particular empirical paper performed a reproducible audit.

24 Position relative to MOND and dark-matter programs

Tau Core should not be framed as anti-MOND or anti-dark-matter. A safer and more accurate position is that it tries to supply a third-level language in which MOND-like phenomenology and dark-matter-like signals may both be effective readouts of deeper source structure.

For the MOND program, the attractive point is that Tau Core does not begin by assuming that the standard cold-particle halo picture is the whole story. It allows the possibility that galaxy-scale acceleration regularities are effective consequences of a deeper gravitational/readout structure. The limiting point, from a strict MOND perspective, is that Tau Core does not promote MOND itself to final ontology. It treats MOND-like laws, if they appear, as possible intermediate effective rules:

$$\text{MOND-like regularity} \quad \rightsquigarrow \quad \text{effective readout limit of a deeper morphology.}$$

This is not a MOND derivation in this paper. It is a claim-boundary statement: Tau Core may try to explain why a MOND-like scale or relation appears, but it must still derive the scale, recover known limits, and pass controls.

For the dark-matter program, the caution is different. Tau Core should not say prematurely that dark matter is an illusion. Lensing, CMB, BAO, cluster, and structure-growth data provide strong evidence for extra gravitational structure in the standard cosmological inference pipeline. A Tau Core branch would have to ask a more decomposed question:

$$\begin{aligned}\text{dark-sector signal} \sim {} & \text{gravitating content} + \text{baryonic remnants} \\ & + \text{modified response} + \text{readout/path/morphology effects,}\end{aligned}$$

where the equality is schematic, not a model equation. The scientific task is to determine which components are present in which readout channel, and which component, if any, behaves like collisionless matter.

Thus the intended public framing is:

> Tau Core is neither anti-MOND nor anti-dark-matter. It is a readout-level formal language in which MOND-like acceleration laws, dark-matter-like lensing/cosmological signals, and observer-dependent residuals can be separated into source-frozen components and tested against one another.

This framing avoids a victory narrative. It does not say that existing communities were simply wrong. It says that different communities may have identified different projections of a deeper problem, and that the useful question is how to split those projections without endpoint leakage.

25 Comparison with adjacent ideas

Tau Core overlaps in spirit with several families of ideas, but it should not be confused with them.

Relational time. Relational-time approaches show how dynamics can be recovered through correlations. Tau Core agrees that external time need not be primitive, but it places this inside a broader readout atlas. Time is one readout among several, not the only reconstruction target.

Shape dynamics and relational configuration. Shape-dynamics-inspired work emphasizes that spacetime structure may be reformulated through different symmetry or relational choices [11]. Tau Core uses this as a cautionary analogy: one must state which structure is primitive and which is recovered. It does not import shape dynamics as a proof.

Quantum reconstruction. Quantum reconstruction programs provide a model for turning operational principles into formal structure. Tau Core's quantum branch must meet a similar standard. It cannot derive Hilbert space merely by naming a quantum readout.

Holography and error correction. Holographic quantum error correction shows that bulk-like structure can be encoded and recovered nontrivially from boundary data [1]. Tau Core treats this as an analogy for readout/reconstruction discipline, not as evidence for the Tau ontology.

26 What would count as progress

The framework becomes stronger only if abstract readout language is converted into bounded tasks. Examples include:

(1) deriving a non-circular parent response law ρ_τ;

(2) defining a source-realizable morphological body M_τ;

(3) proving that a readout map has a quotient-safe null policy;

(4) deriving a known physical limit, such as proper time or Newtonian gravity, as a controlled readout limit;

(5) producing source-frozen empirical rules that survive held-out tests;

(6) preserving negative results by demoting failed kernels instead of retuning them after endpoint inspection.

For the next version, the requested progress should be narrower than another general foundation layer. A single worked branch is the correct target: observer-time, finite readout-time, or another comparably small branch. That branch must choose one mathematical type for ρ_τ, state why that type is admissible for the branch, recover at least one known limit, and freeze at least one exclusion rule before any endpoint inspection. The purpose is not to solve dark matter, quantum mechanics, general relativity, and cosmology at once, but to demonstrate that the readout package can function as a disciplined calculus in one minimal case.

The weakest useful form of progress is a toy model that exposes a blocker. A failed theorem can be valuable if it shows which extra principle is needed. For example, local convexity or a positive Hessian may not force global exposed-face duality; finite contextual grammars may not force Hilbert space; and source stability may not force a unique source basis. Such negative results help keep the framework honest.

The phrase "parent response law" is intentionally not fixed to one mathematical type in this foundation paper. Candidate types include:

variational law	ρ_τ selected by an action or extremal principle,
constraint law	ρ_τ selected by admissibility or closure equations,
algebraic closure	ρ_τ fixed by source/readout consistency,
probabilistic kernel	ρ_τ given by an endpoint-blind transition kernel,
fixed point	ρ_τ selected as a stable solution of a response operator,
functorial rule	ρ_τ induced by structure-preserving source maps.

No choice among these is made here. A later branch must declare which type it uses before claiming a derived physical law.

This is the largest remaining weakness of the foundation layer. The abstract symbol ρ_τ is useful only as bookkeeping until at least one branch fixes its mathematical type and carries a chain

$$\text{source} \to \rho_\tau \to U_i^{M_\tau} \to \text{known limit or frozen falsifiable consequence}$$

without endpoint leakage. Foundation Paper I intentionally stops before that choice; a subsequent worked branch cannot.

27 What would count as failure

A reviewer should not have to infer the failure conditions. At the foundation level, Tau Core would be weakened or demoted if the following failure modes persist after reasonable attempts to repair definitions:

(F1) **No source-realizable parent object.** If no non-circular parent-side source class can be specified for a branch, the branch remains vocabulary rather than physics.

(F2) **No quotient-safe readout.** If declared null-equivalent parent representatives produce different observables, the readout is internally inconsistent.

(F3) **No known-limit recovery.** If a branch cannot recover the relevant standard limit, such as proper time, Newtonian gravity, Born probabilities, or ΛCDM behavior where appropriate, it should not be promoted.

(F4) **Endpoint leakage.** If the source class, kernel family, or amplitude rule is selected using the residual it later explains, the result is a fit, not a Tau Core derivation.

(F5) **Uncontrolled rescue.** If every failure is repaired by adding an unconstrained projection channel after endpoint inspection, the framework loses scientific content.

These are not merely methodological preferences. They are the conditions under which the present foundation language can fail to become a physical research program. A negative result satisfying these conditions should be preserved as a real demotion, not hidden by changing terminology.

For later branches, demotion should be recorded explicitly. The minimal status labels have the following strict meanings:

Status	Strict meaning
TOY-ONLY	no physical codomain, units, or operational observable supplied
PHENOMENOLOGICAL-ONLY	fits or organizes known data but lacks parent-source derivation
BLOCKED	missing known-limit recovery, quotient-safety, or source-realizability
REJECTED	frozen prediction fails holdout/control, or endpoint leakage is found

Forbidden rescue operations include post-hoc kernel exchange, endpoint-selected morphological parameters, retrofitted path factors, and covariance or null policy changes made after endpoint inspection. A promoted branch must provide at least one frozen numerical value, relation, exclusion, or null prediction that could have failed.

Minimal failed-branch example. Let a toy branch declare a null policy $n \sim n'$ and a readout

$$R(x, n) = x.$$

If the branch later adds a correction $A\,n$ to explain an endpoint residual, then it has used a declared null coordinate as a physical observable. The branch fails quotient safety. If A or the choice of n-dependence was selected after inspecting the endpoint residual, it also fails endpoint blindness. Such a branch is not a Tau Core derivation; it is demoted to REJECTED or, if no endpoint was used, at best PHENOMENOLOGICAL-ONLY. This example is deliberately trivial: its purpose is to show that the framework can say “no” before any exotic physics is invoked.

28 Likely objections

A foundation framework should be written so that criticism has handles. We therefore list several likely objections and the intended answer.

Objection 1: this is only a change of language. This objection is serious. The framework is only useful if readout language forces new constraints. The minimum non-linguistic constraints are endpoint blindness, null/quotient safety, source-overlap discipline, known-limit recovery, and explicit failure conditions. Without these, Tau Core would be a vocabulary rather than a research program.

Objection 2: the parent response is unobservable. The parent response is not directly observed by assumption. But this alone is not fatal; many theoretical objects are inferred through constrained effects. The relevant question is whether a declared readout rule produces predictions or exclusions that could have failed. If no such rule is produced, the parent response remains a metaphysical placeholder.

Objection 3: the framework can explain anything. This is the most important danger. Tau Core must therefore preserve negative results. A failed frozen kernel should be demoted, not retuned after looking at the residual. A wrong readout family should remain wrong. A source class that cannot be independently justified should not be promoted. If the framework always rescues itself by adding another projection, it fails as science.

Objection 4: atemporal response sounds like a block universe. The difference is the block-readout relation (3). A block universe treats a 4D history as the final object. Tau Core treats a 4D history as one possible readout. The distinction matters only if other readouts impose additional source-frozen constraints. If they do not, Tau Core collapses back into ordinary effective description.

Objection 5: no physical law is derived. Correct. Foundation Paper I does not derive a physical law. It defines the objects a later derivation must use. A future physical law would require a parent action or closure principle, a domain, a null policy, variation support, units, calibration, and known limits.

29 Structural predictions rather than anomaly corrections

The ontological question is whether the Tau Core language can ever yield a prediction that is not merely a correction to an anomaly in an existing framework such as MOND, ΛCDM, GR, or quantum theory. The answer in Foundation Paper I is conditional.

A Tau Core prediction is not counted as structurally original merely because it adds a residual term to a baseline equation. It would count only if a source-frozen parent/morphology rule implies a constraint before the target endpoint is inspected. The schematic form is

$$\mathrm{Struct}_\tau(s, M_\tau) = 0 \quad \Longrightarrow \quad \mathrm{Constraint}(U_i^{M_\tau}, U_j^{M_\tau}, N_i, N_j),$$

or an exclusion:

$$\mathrm{Struct}_\tau(s, M_\tau) \quad \Longrightarrow \quad \text{readout pattern } P \text{ is inadmissible.}$$

Examples of the relevant kind of constraint include:

- a compatibility condition forcing two readouts to co-vary;
- a null/quotient rule excluding an apparent correction as double counting;
- a basin-transition invariant restricting which readout transitions can appear as stable 4D events;
- a rank, connectivity, or frame-gluing condition that must hold before a Hilbert-like or metric-like representation is allowed;
- a frozen source rule that predicts a missing, forbidden, or correlated feature in a holdout channel.

Such a prediction would be Tau Core-specific only if it is fixed from the parent source, morphological body, null policy, and readout compatibility rules before endpoint inspection. If the same structure is chosen because a known baseline leaves a residual, it is an anomaly-motivated correction rather than a parent-morphological prediction. This paper supplies the criterion; it does not yet supply a validated physical example.

A concrete exclusion at framework level. The framework does forbid one nontrivial modeling move. Suppose two proposed sector corrections are written as

$$\Delta O_i = A_i\, q, \qquad \Delta O_j = A_j\, q,$$

where q is the same source-side coordinate or proxy. Unless the branch declares a source-side split $q = (q_i, q_j)$, an orthogonality/covariance rule, or a shared-parent covariance model before endpoint inspection, the two terms cannot be treated as independent evidence or added as independent corrections. They are the same source degree of freedom read twice. Thus Tau Core forbids "same-source double counting" as an admissible explanation. This is not a new physical prediction, but it is a structural exclusion that goes beyond renaming a residual.

30 Minimal mathematical maturity checklist

Before a branch of Tau Core should be treated as more than a toy model, it should satisfy at least the following checklist.

Requirement	Meaning
Source declaration	The parent-side source or proxy is declared before endpoint scoring.
Readout codomain	The observable space $\mathcal{O}_i$ is specified.
Null policy	Gauge, redundancy, and protected-null directions are declared.
Quotient safety	Equivalent parent representatives give the same readout.
Relift policy	The branch states whether and how J_i is defined.
Known limit	The readout recovers a standard limit where required.
Failure rule	Failed controls cause demotion, not post-hoc retuning.

This checklist is intentionally weaker than a full physical theory. It is also stronger than a metaphor. A branch that cannot fill these rows should not be advertised as a physical result.

31 The Roman foundation sequence

This paper is the first member of a separate Roman-numbered foundation series. It should not be confused with the Arabic-numbered empirical/protocol sequence. The intended division is:

Paper 1–18	bounded empirical or protocol packages,
Foundation Paper I, II, ...	conceptual and mathematical foundation papers.

The proposed next papers are:

- **Foundation Paper I**: atemporal morphological readout; framework only, with no physical validation claim.
- **Foundation Paper II: A Finite Readout-Time Branch**: a worked branch from source declaration to effective readout-time ordering. It should fix the mathematical type of ρ_τ, derive a proper-time or readout-time known limit in a minimal case, and predeclare at least one exclusion such as same-source double counting.
- **Foundation Paper III**: quantum readout; no unconditional Hilbert, Born-rule, or QFT derivation claim.
- **Foundation Paper IV**: hidden defects and dark sectors; no dark-matter or dark-energy replacement claim.
- **Foundation Paper V**: observer-time and cosmology; no Hubble-tension or ΛCDM replacement claim.
- **Foundation Paper VI**: flavor and neutrino readout; no neutrino-anomaly or Standard Model claim.

The next foundation manuscript should not merely add another general framework layer. It should contain at least one worked branch of the form

$$\text{source} \to \rho_\tau \to U_i^{M_\tau} \to \text{known limit or frozen falsifiable consequence.}$$

The purpose of this split is defensive in the good sense. A foundation paper can state a formal conjecture without pretending to be a data paper. A data paper can test a narrow rule without requiring acceptance of the full ontology. This separation makes both kinds of work easier to criticize.

32 A compact formal summary

The framework can be summarized as the following object package:

$$\mathfrak{T} = (\mathcal{S}, \mathfrak{R}_\tau, \rho_\tau, M_\tau, \{U_i\}, \{N_i\}, \{J_i\}).$$

Here $\mathcal{S}$ is a seed/source domain, $\mathfrak{R}_\tau$ is a response domain, ρ_τ is the endpoint-blind response map or relation, M_τ is the morphological body, U_i are readouts, N_i are null policies, and J_i are optional relifts.

The minimal consistency conditions are:

$$\begin{aligned}
&s \text{ is fixed without endpoint leakage,} && (12)\\
&r = \rho_\tau(s) \text{ is defined on the declared source domain,} && (13)\\
&rN_ir' \Rightarrow U_i(r) = U_i(r'), && (14)\\
&\omega_i(r) = 0 \Longleftrightarrow J_iU_i(r)N_ir, && (15)\\
&h_i(r) = J_iU_i(r) - r \text{ is only a chart-level representative if defined,} && (16)\\
&\text{known physical limits are recovered before physical promotion.} && (17)
\end{aligned}$$

The strongest claim of this paper is that $\mathfrak{T}$ is an internally organized starting package for a readout-based research program. This is not a formal consistency proof of all possible Tau Core branches, and it is not a proof that Nature uses such a package.

33 What is not proved

For clarity, we list the main non-results.

- The paper does not prove that the Tau substrate exists physically.
- The paper does not provide a parent action.
- The paper does not derive physical time.
- The paper does not derive general relativity.
- The paper does not derive quantum mechanics, the Born rule, collapse, or QFT.
- The paper does not derive the Standard Model.
- The paper does not explain neutrino physics.
- The paper does not eliminate dark matter or dark energy.
- The paper does not validate any empirical Tau Core claim.

These limitations are not footnotes. They are part of the claim. A foundation paper that hides its blockers is not useful. The goal is to make the blockers visible enough that later work can attack them one at a time.

34 Conclusion

Tau Core is presented here as an atemporal morphological readout framework. The basic proposal is that observed physical dynamics may be sector readouts of an endpoint-blind parent response:

$$s \mapsto \rho_\tau(s) \mapsto U_i^{M_\tau}(\rho_\tau(s)).$$

In this reading, a 4D spacetime block is not the final ontology; it is one readout:

$$M_{4D} = U_{4D}^{M_\tau}(\rho_\tau(s)).$$

Physical time is likewise a readout channel, not the meaning of τ itself.

The paper's contribution is therefore not a new empirical result. It is a bounded vocabulary: parent seed, endpoint-blind response, morphological body, sector readout, relift, hidden defect, observer-time readout, and claim ladder. This vocabulary lets later work ask sharper questions. Which readouts are source-realizable? Which defects survive the quotient? Which known limits are recovered? Which source-frozen kernels survive held-out data? Which candidate branches fail?

The strongest honest summary is:

> Tau Core is not offered here as a completed theory. It is offered as a formal research program in which time evolution, spacetime fields, quantum state, gravity, and dark-sector language can be re-expressed as readout problems over a deeper atemporal morphological response.

If that re-expression leads only to flexible terminology, it should be discarded. If it leads to frozen source rules, recovered limits, and failed or successful held-out tests, then the framework becomes a scientific program. This paper is the first step: it states the program in a form that can be criticized.

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